

# Autonomous Wildfire Early Warning and Pre-Suppression Using Edge-Fog-Cloud Architecture in Indian Forest Ecosystems

Domain: IoT Systems / Edge Computing / Disaster Management / Forest Fire Prevention

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## Abstract

Wildfire detection and management in India has significantly evolved from traditional ground-based patrolling to satellite-integrated cloud-processing systems. While this transition has improved coverage and response accuracy, it introduces a critical structural bottleneck: end-to-end latency. Every observation, prediction, and response command must travel from sensor to satellite, from satellite to cloud, and from cloud back to ground — a pipeline that can span several minutes. In the context of wildfire, where fire spread is non-linear and early minutes are disproportionately critical, this delay is operationally significant.

This paper proposes the design of an **Autonomous Wildfire Early Warning and Pre-Suppression System** built on a three-tier **Edge-Fog-Cloud (EFC) architecture**, specifically tailored for Indian forest ecosystems and the operational structure of Indian forest management agencies. The fog layer — deployed as distributed, locally-powered intelligent nodes within the forest — acts as a real-time decision-making layer that operates independently of cloud availability, triggering pre-suppression responses and issuing ground alerts within seconds of detecting critical conditions. This system does not replace the satellite-cloud pipeline but augments it, addressing the latency gap at the critical early intervention window.

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## 1. Introduction

India holds approximately **71.6 million hectares** of forest cover, accounting for roughly 21.7% of its total geographical area (Forest Survey of India, 2021). These forests are not only ecological assets but also home to significant biodiversity, tribal communities, and carbon sinks of global importance. Yet, they are increasingly vulnerable to wildfires — a threat that has intensified due to climate change, rising temperatures, extended dry seasons, and land-use pressure.

According to the **Forest Survey of India (FSI)** and data from the **National Remote Sensing Centre (NRSC)**, India records tens of thousands of forest fire incidents annually. States like Uttarakhand, Odisha, Chhattisgarh, Andhra Pradesh, and Madhya Pradesh are among the most fire-prone. The **2021 Uttarakhand wildfire** and **recurring fire seasons in the Eastern Ghats** have demonstrated the scale of destruction that delayed detection and response can cause.

The dominant current approach relies heavily on satellite remote sensing — specifically the **MODIS (Moderate Resolution Imaging Spectroradiometer)** and **VIIRS (Visible Infrared Imaging Radiometer Suite)** sensors operated under NASA's FIRMS (Fire Information for Resource Management System), which is integrated into India's own **FAST (Forest Fire Alert System)** managed by FSI and NRSC. While this system provides national-scale monitoring, it operates on fixed satellite pass intervals and inherits the round-trip latency of cloud-side processing.

This paper argues that the next meaningful step in India's wildfire management technology stack is not replacing this satellite-cloud system but introducing a **complementary low-latency intelligent layer** — the **fog computing layer** — that sits between field sensors and the cloud, enabling autonomous early action before the satellite even completes its next pass.

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## 2. Problem Statement

### 2.1 The Latency Problem in Current Systems

The current satellite-cloud pipeline for wildfire detection operates as follows:

**Field Observation → Satellite Capture → Cloud Processing → Alert Generation → Control Center → Decision → Ground Action**

Each stage introduces delay:

| Stage                              | Approximate Delay |
|------------------------------------|-------------------|
| Satellite revisit interval (MODIS) | 1–2 times per day |

|                                       |                              |
|---------------------------------------|------------------------------|
| Satellite revisit interval (VIIRS)    | ~12 hours                    |
| Cloud processing and alert generation | 5–30 minutes                 |
| Alert reaching local control center   | Variable (network dependent) |
| Human decision + dispatch             | 30 minutes to several hours  |
| Ground team reaching site             | Hours (terrain dependent)    |

**Total realistic delay: 1 to 12+ hours** from early ignition indicators to any ground action.

Meanwhile, wildfire spread is governed by the **fire triangle** (fuel, oxygen, heat) and environmental conditions. Under moderate wind and dry conditions, a fire can double its area every 10–20 minutes. An hour of unresponded early warning can translate to hundreds of acres of spread.

## 2.2 Absence of Local Autonomous Intelligence

The current system depends entirely on human decision-making at the control center level. While automation exists in the cloud (threshold-based alerts), there is **no autonomous decision-making layer that is physically close to the event** and can act without waiting for human instruction or cloud response.

This is the core gap: the system is **fast at seeing** but **slow at acting**.

## 2.3 Connectivity and Cloud Dependency Risks

Forest terrain in India, particularly in the Western Ghats, Northeast, and Himalayan zones, experiences **intermittent satellite connectivity and unreliable backhaul links**. A purely cloud-dependent architecture fails precisely when connectivity is degraded — often during storms, which are correlated with high fire-risk dry spells that follow.

## 2.4 Scale and Geographic Distribution

India's forest cover is distributed across **diverse terrain, climatic zones, and administrative structures**. A centralized cloud-only model struggles to account for hyper-local microclimatic conditions that influence fire risk — conditions that can vary drastically within even a few kilometers of forest.

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## 3. Review of Existing Systems

### 3.1 Satellite-Based Systems (FIRMS / FAST)

NASA FIRMS provides near-real-time fire data globally using MODIS and VIIRS sensors. India's FSI and NRSC have integrated this into the **Forest Fire Alert System (FAST)**, which sends SMS/email alerts to range officers and divisional forest officers based on satellite-detected hotspots.

#### Strengths:

- Wide area coverage
- No ground infrastructure needed
- Continuous improvement in resolution and frequency

#### Limitations:

- Fixed revisit intervals (not continuous)
- Cannot detect pre-ignition conditions (only active fire)
- Dependent on cloud-free line of sight
- High latency between condition and alert

### 3.2 IoT-Based Forest Monitoring (Research Stage)

Several academic and pilot studies have explored IoT sensor networks for forest fire detection. Systems using temperature, humidity, smoke, and CO sensors transmitted over LoRa or ZigBee to a base station have been proposed and tested in limited deployments.

#### Limitations of existing IoT proposals:

- Most are data collection systems, not decision-making systems
- Rely on cloud for all processing
- Lack coordination between nodes
- No autonomous pre-suppression logic
- Not designed for India-specific terrain, power, or regulatory context

### 3.3 Drone-Based Surveillance (Pilot Level)

Some state forest departments (notably Uttarakhand and Telangana) have piloted drone-based surveillance for fire detection. These are operated manually or on fixed patrol routes.

#### Limitations:

- Reactive, not predictive
- Human-operated
- Limited flight time and coverage
- Not integrated into a broader decision system

### 3.4 Summary of the Gap

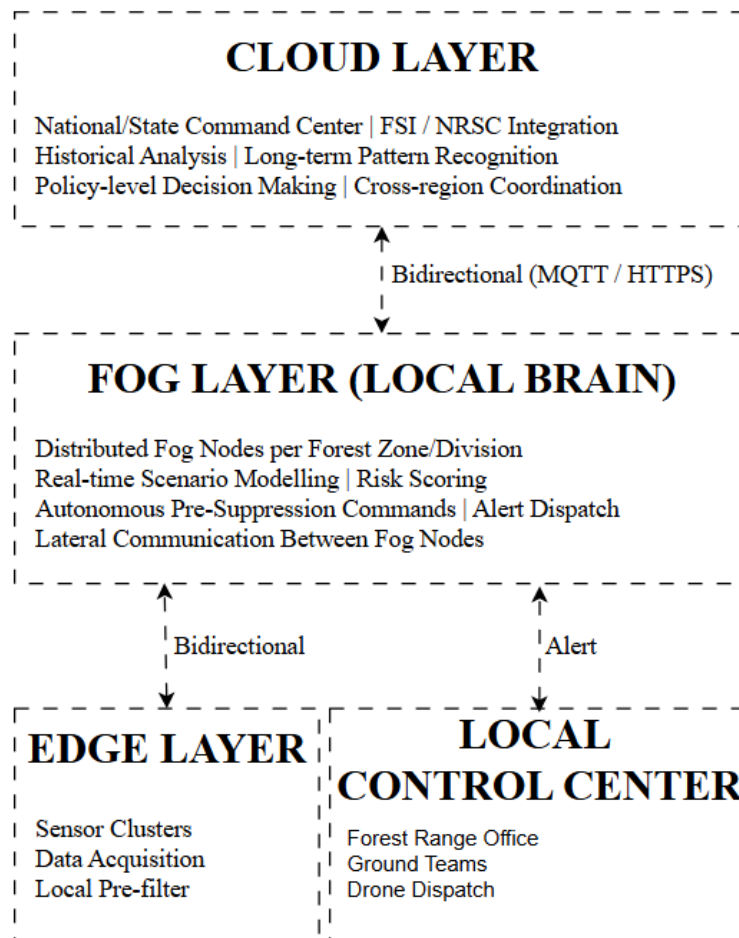
| Capability                          | Satellite-Cloud | Existing IoT Proposals | Proposed EFC System |
|-------------------------------------|-----------------|------------------------|---------------------|
| Real-time field sensing             | ✗               | ✓                      | ✓                   |
| Pre-ignition detection              | ✗               | Partial                | ✓                   |
| Low-latency local response          | ✗               | ✗                      | ✓                   |
| Autonomous decision-making          | ✗               | ✗                      | ✓                   |
| Cloud offloading (non-critical)     | ✓               | Partial                | ✓                   |
| Offline/disconnected operation      | ✓               | ✗                      | ✓                   |
| Coordinated multi-node intelligence | ✗               | ✗                      | ✓                   |

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## 4. Proposed System: Edge-Fog-Cloud Architecture for Wildfire Management

### 4.1 Architecture Overview

The proposed system operates on three tiers:



## 4.2 Edge Layer — Field Sensor Clusters

Each **Edge Node** is a weatherproof sensor cluster deployed at strategic points across the forest (based on fire risk mapping). They continuously sample environmental parameters and transmit to the nearest fog node.

### 4.2.1 Sensor Stack (per Edge Node)

| Sensor                          | Parameter Measured              | Relevance to Fire Risk          |
|---------------------------------|---------------------------------|---------------------------------|
| DHT22 / SHT40                   | Temperature + Relative Humidity | Dry, hot conditions = high risk |
| Anemometer + Wind Vane          | Wind speed + direction          | Determines fire spread vector   |
| Capacitive Soil Moisture Sensor | Soil moisture content           | Dry soil = high fuel load       |

|                             |                          |                             |
|-----------------------------|--------------------------|-----------------------------|
| MQ-2 / MQ-7 Gas Sensor      | Smoke, CO, LPG           | Early combustion indicator  |
| FLIR Lepton / MLX90640      | Infrared thermal imaging | Surface temperature anomaly |
| BH1750 / TSL2591            | Ambient light intensity  | Time-of-day context         |
| Rain Gauge (Tipping Bucket) | Precipitation            | Natural suppression factor  |
| GPS Module (NEO-8M)         | Geo-location             | Spatial mapping of events   |

#### 4.2.2 Edge Node Communication

- **Protocol:** LoRa (Long Range, Low Power) — 868 MHz band in India
- **Range:** 2–15 km line-of-sight in open terrain; 1–5 km in forested areas
- **Data rate:** Low (sensor telemetry is small payloads — this is appropriate for LoRa)
- **Topology:** Star topology to nearest Fog Node; mesh fallback between edge nodes

#### 4.2.3 Power Design

- **Primary:** Solar panel (10W–20W monocrystalline) + LiFePO<sub>4</sub> battery bank (sufficient for 5–7 days of overcast operation)
- **Backup:** Supercapacitor bank for surge protection
- **Rationale:** Eliminates grid dependency entirely; designed for deployment in remote forest zones

### 4.3 Fog Layer — The Local Brain

This is the **core innovation** of the proposed system. Each fog node is an intelligent computing unit deployed at the forest range/beat level — physically close to the field, not at the district control center.

#### 4.3.1 What the Fog Node Is

A fog node is a **ruggedized edge server** — not a cloud server, not a microcontroller. It sits in the middle of the compute hierarchy.

**Candidate hardware (mid-level spec):**

- NVIDIA Jetson Orin NX (for AI inference capability)

- Or: Industrial-grade ARM server (e.g., Advantech EPC-R3720) for non-AI-intensive deployments
- Redundant storage (eMMC + SD backup)
- Cellular modem (4G LTE + fallback 2G) for cloud uplink
- Local LoRa gateway for edge communication
- Housed in IP67-rated enclosure

#### 4.3.2 What the Fog Node Does

**Step 1 — Continuous Data Ingestion** Receives time-series sensor data from all edge nodes in its zone via LoRa gateway. Maintains a rolling local data buffer.

**Step 2 — Scenario Construction** The fog node doesn't just threshold-check individual sensors. It builds a **composite risk scenario** by correlating multiple parameters:

**Risk Scenario = f(Temperature, Humidity, Wind Speed, Wind Direction,  
Soil Moisture, Gas Levels, Thermal Anomaly,  
Time-of-day, Season, Historical fire data for zone)**

This is implemented as a **multi-parameter weighted risk scoring model**, initially rule-based (transparent, auditable) and extensible to lightweight ML models (e.g., Random Forest, TinyML) in later phases.

**Step 3 — Risk Classification** The scenario is classified into one of four states:

| State  | Description          | Action Triggered                                             |
|--------|----------------------|--------------------------------------------------------------|
| GREEN  | Normal               | Log and report to cloud periodically                         |
| YELLOW | Elevated Risk        | Increase sampling rate; alert control center                 |
| ORANGE | Imminent Risk        | Pre-suppression activation; human alert + cloud notification |
| RED    | Active Fire Detected | Full autonomous response + emergency escalation              |

**Step 4 — Autonomous Pre-Suppression (ORANGE/RED)** Within the fog node's **safe action envelope** (defined actions that are pre-authorized and bounded), it can directly command:

- Activation of fixed **mist/sprinkler systems** at perimeter fire-break points
- Deployment command to **pre-positioned autonomous drones** (if available)
- Activation of **acoustic deterrent systems** to drive wildlife away from risk zone

- Triggering of **physical firebreak actuators** (motorized clearing barriers, if installed)

All autonomous actions are:

- Logged with timestamp and sensor justification
- Bounded to pre-defined safe envelopes (cannot exceed authorized scope)
- Immediately reported to human control center with reason

### Step 5 — Dual Parallel Reporting

Simultaneously with autonomous action:

Report A → Local Control Center (immediate, short-format alert)

Contains: Zone ID, Risk Level, Key parameters, Actions taken

Report B → Cloud Layer (full data packet)

Contains: Raw sensor time-series + Scenario analysis +

Decision log + Actions taken + Request for cloud-level analysis

**Step 6 — Receiving Cloud Commands** If the cloud layer sends override or refined commands based on its broader analysis, the fog node adjusts its behavior accordingly. The fog node always defers to cloud commands when connectivity is available — it only operates autonomously when cloud is unreachable or during the response latency window.

### 4.3.3 Fog Node Clustering and Lateral Communication

Multiple fog nodes are deployed across forest divisions. They communicate laterally via:

- **Primary:** 4G LTE mesh (where available)
- **Fallback:** Long-range WiFi (802.11ah — 900 MHz sub-GHz band) or licensed RF links

Lateral communication enables:

- **Fire spread prediction:** If Node A detects ignition and wind is blowing toward Node B's zone, Node A proactively alerts Node B to enter pre-suppression state
- **Shared situational awareness:** All nodes maintain a synchronized regional risk map
- **Redundancy:** If one fog node fails, adjacent nodes expand their coverage zone

### 4.3.4 Fog Node Power Design

- **Primary:** Solar + battery (larger capacity than edge nodes — 100Ah+ LiFePO4 bank)
- **Backup:** Small diesel/petrol generator for extended outage
- **Rationale:** Fog nodes are semi-permanent installations at range offices or designated forest posts; limited grid access is acceptable; full off-grid preferred

## 4.4 Cloud Layer

The cloud layer serves as the **strategic intelligence and long-term management brain**. It does not replace operational fog decisions but provides context that fog nodes cannot generate alone.

### Functions:

- Ingests fog node reports and raw sensor data
- Runs **historical comparison models** (current conditions vs. same season in past years)
- Maintains **national-level fire risk maps** updated in near-real-time
- Interfaces with **FSI/NRSC satellite systems** — correlating satellite hotspot data with fog-level sensor data
- Sends **advisory commands** to fog nodes (e.g., recalibrate risk thresholds for incoming heat wave)
- Generates **post-event analysis reports** for forest department records and policy use
- Hosts the **administrative dashboard** for district/state/national forest officers

**Cloud infrastructure:** Can be hosted on NIC Cloud (Government of India's MeghRaj) or state data centers, aligning with data sovereignty requirements.

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## 5. System Workflow — End to End

### 5.1 Normal Operation (GREEN State)

Edge sensors → LoRa → Fog Node

Fog Node: Rolling data collection, scenario scoring, GREEN state confirmed

Fog Node → Cloud: Periodic telemetry batch (every 15–30 min)

Cloud: Updates national risk map, no action required

## 5.2 Elevated Risk (YELLOW State)

Fog Node detects rising temperature + dropping humidity + increasing wind speed

Risk Score crosses YELLOW threshold

Fog Node: Increases sampling rate (edge nodes polled every 30 sec instead of 5 min)

Fog Node → Control Center: "Zone 4B: Elevated risk. Monitor."

Fog Node → Cloud: Real-time streaming begins

Fog Node → Adjacent Fog Nodes: "Zone 4B elevated — downstream zones stand by"

## 5.3 Imminent Risk / Autonomous Action (ORANGE State)

Gas sensor detects CO spike. Thermal camera shows surface temp anomaly.

Soil moisture critically low. Wind direction: toward fuel-dense zone.

Risk Score crosses ORANGE threshold.

Fog Node (within seconds):

- ACTION: Activates perimeter mist systems at fire-break points
- ACTION: Sends deployment command to nearest drone station
- ALERT → Control Center: Full scenario brief + actions taken
- REPORT → Cloud: Complete data packet + decision log

Control Center receives alert:

- Reviews fog node assessment
- Dispatches ground team (with fog-generated zone map)
- Can override fog actions if needed

Cloud receives report:

- Cross-references satellite data
- Runs predictive spread model
- Sends refined tactical advisory back to fog node

## 5.4 Active Fire (RED State)

Fog Node confirms active fire (thermal + gas + visual)

Full escalation:

- All pre-suppression systems at maximum
  - Emergency alert to control center + state fire department
  - Fog lateral broadcast: all neighboring nodes escalate to ORANGE
  - Cloud: Emergency coordination mode activated
  - Integration with NDMA (National Disaster Management Authority) protocols
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# 6. Innovations and Technical Contributions

## 6.1 Fog-Level Autonomous Decision Making for Wildfire

No existing wildfire management system in India (or in published literature for Indian forest conditions) implements an autonomous decision-making fog layer. The novelty is not fog computing itself (which is established) but its **specific application to wildfire pre-suppression with a bounded autonomous action framework**.

## 6.2 Multi-Parameter Composite Risk Scoring

Moving beyond single-parameter threshold alerts (e.g., "temperature > X = alert") to a **composite, weighted, time-aware scenario model** that accounts for the interplay of environmental factors is a more robust and lower false-positive approach.

## 6.3 Lateral Fog Node Coordination for Fire Spread Prediction

The fog cluster's ability to **predictively alert downstream nodes** based on wind direction and fire behavior modeling — before fire actually reaches those zones — is a proactive capability absent from current systems.

## 6.4 Offline-Resilient Architecture

The system is explicitly designed to function at fog level even with **zero cloud or satellite connectivity** — a critical requirement for Indian forest terrain where connectivity is unreliable.

## 6.5 Integration Layer Between IoT and Satellite Systems

The system creates a **data bridge** between ground-truth IoT sensor data and satellite-cloud data — enabling the cloud to validate satellite detections against sensor-level ground truth, reducing false positives from satellite systems (e.g., industrial heat sources being misidentified as wildfires).

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# 7. Effectiveness and Impact Assessment

### 7.1 Reduction in Response Time

| System                          | Detection to First Action      |
|---------------------------------|--------------------------------|
| Current Satellite-Cloud         | 1–12+ hours                    |
| Proposed EFC System (fog level) | 30 seconds to 3 minutes        |
| EFC + Cloud Coordination        | 3–10 minutes for full response |

This represents an order-of-magnitude improvement in initial response time.

### 7.2 Fire Area Reduction Potential

Using standard fire spread models (Rothermel model, adapted for Indian forest types), a 1-hour reduction in response time under moderate wind conditions (15 km/h) can prevent fire spread across an estimated **10–50 additional hectares** per incident, depending on fuel load and terrain.

At India's current incident rate, this can translate to **thousands of hectares of forest preserved annually**.

### 7.3 Reduction in Human Risk

By enabling autonomous early pre-suppression before ground teams arrive, the system reduces the conditions ground teams face on arrival — **lower flame height**,

**contained perimeter, reduced smoke** — directly improving operational safety for forest firefighters.

## 7.4 Cloud Efficiency Improvement

By filtering and pre-processing data at the fog level, the system reduces the volume of raw sensor data that must be transmitted to and processed by the cloud — improving cloud pipeline efficiency and reducing bandwidth cost.

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# 8. Addressing Implementation Challenges

## 8.1 Power in Remote Forest Areas

**Challenge:** No grid power in deep forest zones.

**Solution:** Each edge and fog node is fully solar-powered with battery backup sized for 5–7 days of zero-solar operation (monsoon/heavy overcast periods). LiFePO<sub>4</sub> chemistry is chosen for longevity (2000+ charge cycles), thermal stability in high-temperature environments, and safety (no thermal runaway risk unlike Li-ion). Energy harvesting from wind (small turbines) can supplement solar at exposed ridgeline positions.

## 8.2 Connectivity in Forest Terrain

**Challenge:** Cellular and satellite connectivity is unreliable in dense forest / hilly terrain.

**Solution:** Multi-layer communication stack:

- **Edge to Fog:** LoRa (no infrastructure, long range, low power) — works without internet
- **Fog to Cloud:** 4G LTE primary; VSAT secondary for deep zones; **the system is designed to function fully at fog level with zero cloud connectivity**
- **Fog to Fog:** Sub-GHz WiFi or licensed RF mesh — independent of cellular infrastructure

## 8.3 Autonomous Decision Governance

**Challenge:** Who is legally and operationally responsible if an autonomous system takes an incorrect action?

**Solution — Bounded Autonomy Framework:**

- All autonomous actions are within a **pre-defined, government-approved action envelope**
- Actions are limited to **safe, reversible interventions** (mist systems, acoustic deterrents, alerts)
- Every autonomous action generates an **immutable log entry** with sensor justification, timestamp, and action taken
- Human override is always available and takes precedence
- The system is positioned as a **decision support and early responder**, not a replacement for human authority
- Governance framework to be co-designed with Forest Department operational protocols

## 8.4 False Positive Management

**Challenge:** Sensor faults or environmental anomalies could trigger false alarms.

**Solution:**

- Multi-sensor confirmation requirement: **at least 3 independent sensor parameters** must cross threshold simultaneously to trigger ORANGE/RED state
- Sensor self-diagnostic routines running continuously
- Fog node maintains **sensor health scoring** — degraded sensors are down-weighted in risk calculation
- Cloud-level historical comparison helps identify pattern anomalies vs. genuine events

## 8.5 Hardware Durability in Forest Environment

**Challenge:** Temperature extremes, humidity, insects, wildlife interference.

**Solution:**

- All enclosures: IP67 (dust-tight, water-immersion rated) minimum
- Operating temperature range: -10°C to +65°C (covers Indian forest range from Himalayan to tropical zones)
- Tamper detection sensors on enclosures
- Anti-corrosion coatings on all exposed metal
- Mesh wildlife guard on ventilation ports

## 8.6 Regulatory Considerations

**Drone deployment:** Subject to DGCA (Directorate General of Civil Aviation) regulations. In the proposed system, drone deployment commands from fog nodes apply only to **pre-authorized, registered drones operating within pre-approved**

**operational zones** under existing Forest Department permissions. This is consistent with DGCA's framework for government/public safety operations.

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## 9. Deployment Strategy

### 9.1 Phased Rollout

#### Phase 1 — Pilot (Year 1)

- Select one high-risk forest division (recommended: Simlipal, Odisha or Corbett, Uttarakhand)
- Deploy 5–10 edge nodes + 1 fog node
- Cloud dashboard integration with FSI/NRSC data feed
- Validate sensor stack, risk model, and communication reliability
- No autonomous action in Phase 1 — fog node in advisory-only mode

#### Phase 2 — Validation (Year 2)

- Expand to 3–5 forest divisions across 2 states
- Enable autonomous pre-suppression in bounded envelope (mist systems only)
- Lateral fog node coordination enabled
- Publish Phase 1 performance data; calibrate risk model with ground truth

#### Phase 3 — Scaled Deployment (Year 3+)

- State-level rollout in 2–3 fire-prone states
- Full autonomous response stack including drone coordination
- Integration with national FSI dashboard
- Model open-sourced for replication by other states

### 9.2 Institutional Integration

The system is designed to work **within existing government structures**, not parallel to them:

- Fog alerts go to existing **Range Forest Officers (RFOs)** — no new hierarchy
  - Cloud dashboard integrates with existing **FSI State of Forest Report** data infrastructure
  - Hardware maintenance tied into existing **Forest Department field staff responsibilities**
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## 10. Limitations and Future Work

### Current Limitations:

- Risk scoring model requires calibration per forest type and climatic zone — a one-size model is insufficient
- Fog hardware cost is non-trivial; economic viability analysis needed for large-scale deployment
- Drone-based pre-suppression is aspirational at this stage — regulatory and logistical groundwork needed
- Long-term sensor calibration drift in field conditions is an open engineering problem

### Future Work:

- Integration of **TinyML models** on edge nodes for on-device anomaly detection
  - **Digital twin** of forest zones at cloud level for simulation-based planning
  - Citizen science integration — forest-adjacent community reporting feeding into fog risk model
  - Interoperability with **NDMA's common alerting protocol (CAP)** for multi-agency coordination
  - Carbon credit quantification for prevented forest loss — creating an economic incentive model for deployment
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## 11. Conclusion

India's wildfire management has matured significantly through satellite-cloud integration, but it carries a structural latency gap that becomes operationally critical during early-stage fire events. This paper has proposed an Edge-Fog-Cloud architecture that introduces a **locally-intelligent, autonomously-acting, offline-resilient fog layer** into the forest fire management stack — designed specifically for Indian forest ecosystems, administrative structures, and terrain constraints.

The system does not displace existing satellite and cloud infrastructure; it augments it with a **sub-minute decision loop** that can initiate pre-suppression and human alerts before satellite systems complete a single revisit cycle. The fog layer's bounded autonomy, multi-sensor risk model, lateral node coordination, and offline resilience collectively address the core limitations of existing approaches.

The result is a system where **detection, decision, and initial action are measured in seconds**, ground teams are better-informed before they arrive, and the cloud operates

on richer, ground-validated data — a meaningful architectural step forward for India's forest fire response capability.

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*Note: This document represents the conceptual and mid-level technical design of the proposed system. Full implementation specifications, hardware procurement lists, deployment cost analysis, and field trial methodology are addressed in the accompanying technical annexure [to be developed].*